

Hefeifei Jiang

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Education

UCR(University of California, Riverside)
M.S. in Robotics

Riverside, CA
Sep. 2025 - Present

Joint-Degree Undergraduate Program, Shaanxi University of Science and
Technology (SUST) & Ulster University
Bachelor of Engineering (First-class degree), Major: Computer Science and Technology

Xi'an, China
Sep. 2020 – Jun. 2024

Publication

Extracting Fingerprint Features Using Autoencoder Networks for Gender
Classification

Applied Sciences
2022

- Y. Qi, M. Qiu, **H. Jiang**, and F. Wang, “Extracting Fingerprint Features Using Autoencoder Networks for Gender Classification,” *Applied Sciences*, 12(19): 10152, 2022. IF = 2.838. doi:10.3390/app121910152.

ConvLSTM Coupled Economics Indicators Quantitative Trading Decision
Model

Symmetry
2022

- Y. Qi, **H. Jiang**, S. Li, and J. Cao, “ConvLSTM Coupled Economics Indicators Quantitative Trading Decision Model,” *Symmetry*, 14(9): 1896, 2022. IF = 2.940. doi:10.3390/sym140918964.

AIOT Digital Twin Intelligent Substation Cloud-based Analysis Platform

Patent

- Y. Qi, **H. Jiang**, X. Jin, J. Ma, M. Zhao, Y. Liu, K. Qian, Y. Wang, Z. Wang, H. Cui, “AIOT Digital Twin Intelligent Substation Cloud-based Analysis Platform,”.

A Method and Apparatus for Matching a Three-dimensional Model of the
Sacrum Based on Deep Learning

Patent

- Y. Qi, **H. Jiang**, F. Wang, J. Yang, “A Method and Apparatus for Matching a Three-dimensional Model of the Sacrum Based on Deep Learning,”.

Research Experience

Traffic Signal Control Using VLM model and Reinforcement Learning

Graduate Researcher, Trustworthy Autonomous Systems Laboratory (TASL),
supervised by Prof. Jiachen Li

Riverside, CA
Oct. 2025 – Present

- Conducting research on *traffic signal control* using reinforcement learning (RL) augmented with vision-language models (VLMs) to handle complex and safety-critical intersection scenarios.
- Designed RL-based signal control policies that incorporate multi-modal inputs (visual perception and semantic reasoning) to improve decision-making under heterogeneous traffic conditions.
- Focused on long-tail and rare-event scenarios, including emergency vehicles (e.g., fire trucks and ambulances), abnormal traffic flows, and non-recurrent congestion patterns.
- Developed evaluation protocols beyond conventional efficiency metrics (e.g., average delay and queue length), introducing sustainability-oriented objectives such as carbon emission reduction and environmentally aware traffic efficiency.
- Validated proposed methods in simulated intersection environments, demonstrating improved overall traffic performance, robustness in edge cases, and enhanced trustworthiness of autonomous traffic control systems.

Quantum Neural Networks for Efficient Multimodal Representation Learning

Remote

Remote Research Intern, supervised by Prof. Juan Zhuang

Jun. 2024 – Dec. 2024

- Conducted research on replacing the image encoder in CLIP with a Quantum Neural Network (QNN) to reduce model size while preserving multimodal representation performance.
- Motivated by the high parameter cost of large multimodal models, explored QNNs as a compact alternative capable of representing high-dimensional visual features with significantly fewer trainable parameters.
- Designed and implemented a QNN-based visual encoder and integrated it into a CLIP-style multimodal framework for comparative evaluation.
- Evaluated the proposed approach on benchmark datasets, demonstrating competitive performance with a substantially reduced parameter budget.
- This project marked my first experience at the intersection of quantum computing and deep learning, deepening my understanding of efficient model design and multimodal learning.

Summer Workshop at National University of Singapore

Singapore

Team Leader, supervised by Prof. Tan Wee Kek

Jun. 2024 – July. 2024

- Built a smart greenhouse monitoring system using Raspberry Pi, ESP32, and multi-modal sensors (temperature, humidity, CO₂, soil moisture, and soil temperature).
- Developed a full-stack web application for real-time environmental monitoring and pest detection in greenhouse settings.
- Implemented data transmission and API services using Flask to stream sensor readings and video feeds from Raspberry Pi to a local server for visualization.
- Applied YOLOv8l on live video streams for potted-plant counting; captured leaf images and used a ResNet-based classifier for pest and disease recognition.
- Coordinated task decomposition and integration in a multinational team; strengthened system-level research execution (hardware–software co-design).

Automated Annotation of Sacrum Bones (with Xi'an Honghui Hospital)

Xi'an, China

Project Leader, supervised by Prof. Yong Qi

Jan. 2022 – Sep. 2023

- Addressed annotation-quality bottlenecks in medical image segmentation by collaborating with Xi'an Honghui Hospital to acquire sacrum bone data.
- Designed a semi-automatic labeling workflow: rough labeling for ROI localization and contrast enhancement via histogram equalization to sharpen boundaries.
- Implemented an OpenCV-based contour recognition pipeline to extract annotated regions and generate refined masks for automatic annotation.
- Proposed a HU-value preservation strategy: restored Hounsfield Unit values from DICOM files by replacing internal sacrum-mask pixels with original HU values to retain bone density cues.
- Prepared a research manuscript (in progress); contributed a practical annotation pipeline and dataset-quality improvement methodology for downstream segmentation training.

Using Autoencoder Networks to Extract Fingerprint Features for Gender Classification

Xi'an, China

Project Leader, supervised by Prof. Yong Qi

Jul. 2021 – Oct. 2022

- Proposed a Dense Dilated Convolution ResNet Autoencoder to extract discriminative gender-related fingerprint representations by replacing standard convolutions with dilated convolutions.
- Evaluated multiple model designs by benchmarking 6 encoder variants and 9 classifier heads on an in-house fingerprint dataset.
- Conducted controlled experiments across different fingers and hand sides; observed that male features were more abundant and ring-finger patterns were more distinctive.
- Applied Class Activation Mapping (CAM) to interpret learned features, highlighting the influence of ring/spiral patterns, branching structures, and linear textures.

Digital Twin Intelligent Grid Physical Fusion Platform

China

Robot Embedded Development

Apr. 2021

- Developed a smart-grid monitoring framework integrating digital-twin concepts with inspection robots and cloud-based management software for alerting and maintenance.

- Built a tracked inspection robot with SLAM-based mapping and multi-sensor deployment for complex terrain operations; integrated obstacle detection and automatic inspection functions.
- Trained and deployed a YOLOv3-based visual recognition module using domain electrical-utility datasets; supported real-time multi-source data monitoring.
- Project outcomes included a software copyright and a provincial-level innovation/entrepreneurship project deployed in multiple substations.

Survey of Sacrum Segmentation Methods

Participant(Paper Writing and Experiment conducted), supervised by Prof. Yong Qi

Xi'an, China

Aug. 2022 – Apr. 2023

- Conducted a systematic survey on sacrum segmentation, motivated by clinical needs in fracture/osteoporosis-related diagnosis and treatment planning.
- Reviewed and summarized deep learning segmentation paradigms and recent advances relevant to sacrum and bone segmentation tasks.
- Co-authored a survey manuscript titled “Survey on Bone Segmentation Methods on Sacrum Osteoarthritis Research: A Deep Learning Approach” (submitted).
- Strengthened research skills in literature synthesis, experimental design planning, and medical imaging methodology evaluation.

Honors & Awards

INTERNATIONAL

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| Feb. 2022 Meritorious Award , Mathematical Contest in Modeling (MCM) | <i>United States</i> |
| Apr. 2021 Honorable Mention , ICPC Asia-East Continent Final | <i>Asia</i> |
| Aug. 2020 Gold Medal , International Festival of Science and Technology | <i>Tunisia</i> |

DOMESTIC

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| Jun. 2024 “Zhi Cheng Zhi Bo” All-Round Excellence Scholarship (Top 10) , University-level Scholarship | <i>China</i> |
| Jun. 2024 5th Place , 4th “Keda” Cup Five-a-Side Football Tournament | <i>China</i> |
| Apr. 2024 Zhou Lian Scholarship (Top 10, #1 by points) , University-level Scholarship | <i>China</i> |
| Aug. 2022 3rd Prize , National College Optoelectronic Design Competition | <i>China</i> |
| Aug. 2022 3rd Prize , National College Computer Design Competition | <i>China</i> |
| May 2022 Ranked 215 / 627 , WeChat Big Data Challenge | <i>China</i> |
| Oct. 2021 1st Prize , 23rd National Robot Championship & 12th International Humanoid Robot Olympics | <i>China</i> |
| Nov. 2021 Provincial-Level Approval , China College Student Innovation and Entrepreneurship Project | <i>China</i> |
| Dec. 2021 Excellence Award , China AI+ Innovation and Entrepreneurship Competition | <i>China</i> |
| Aug. 2021 3rd Prize , National College Computer Design Competition | <i>China</i> |